

Judged Evaluation Report: Automotive Diagnostics

RAG Agent Performance Analysis (Use Case: Mechanics)

1. Executive Judicial Summary

Unlike the previous agriculture dataset (which suffered from retrieval failures and partial hallucinations), **the RAG agent performed exceptionally well on this new automotive diagnostic document (car_final.txt).**

The success rate is 100%. The agent perfectly extracted suspicious noises, correctly identified mechanical failures (wheel bearings, steering racks, clutches), and outputted exact repair protocols without any entity confusion.

2. Detailed Evaluation Registry (6 Questions)

Question 1 : What are the main causes and typical fixes for worn brake pads?

Expected: The main causes are normal wear, cheap pads, and aggressive braking. The typical fix is to replace the brake pads and check the brake rotors.

Agent Prediction: The main causes of worn brake pads include normal wear, using cheap pads, and aggressive braking. The typical fix involves replacing the brake pads and checking the brake rotors.

Judge Verdict: *Well answered. Perfect extraction of both the root causes and the required maintenance steps.*

Question 2 : How can a driver recognize a bad steering rack while driving, and is a garage required for the repair?

Expected: A driver can recognize it by a loose steering feeling, a clunk noise while turning, and an unstable steering feel. Yes, a garage is required to repair or replace the steering rack.

Agent Prediction: While driving, you can recognize a bad steering rack by experiencing a loose steering feeling, hearing a clunk noise while turning, and noticing that the steering may feel unstable. A garage is required for the repair, as it involves specialized tools and expertise.

Judge Verdict: Well answered. The agent successfully listed the exact symptoms and correctly confirmed the need for a professional garage.

Question 3 : What does it mean if you hear a roaring or airplane-like noise that gets louder with speed?

Expected: This sound indicates a bad wheel bearing. It means the wheel bearing is worn out due to wear, water damage, impacts, or water intrusion.

Agent Prediction: If you hear a roaring or airplane-like noise that gets louder with speed, it could indicate a bad wheel bearing. This is a high-risk issue that requires professional garage service to address. The typical fix is to replace the wheel bearing.

Judge Verdict: Well answered. The diagnosis is spot-on (wheel bearing), though it slightly abbreviated the specific list of causes (water damage, etc.) in favor of highlighting the risk level.

Question 4 : What should you do if you hear a rhythmic clunking or wobbling noise and notice wheel wobble that increases with speed?

Expected: This indicates loose lug nuts, which is extremely dangerous. You should tighten the lug nuts immediately.

Agent Prediction: If you hear a rhythmic clunking or wobbling noise and notice wheel wobble that increases with speed, it sounds like your wheel nuts might be loose. This is extremely dangerous and should be addressed immediately. If caught early, you can tighten the lug nuts yourself. However, if the issue persists or you are unsure, it's best to have a professional at a garage tighten them for you.

Judge Verdict: Well answered. Accurate problem identification and perfectly conveyed the severe urgency/danger of the situation.

Question 5 : What are the main causes of engine overheating, and what are the typical fixes?

Expected: The main causes are low coolant, a coolant leak, or old cooling parts. Typical fixes include repairing the leak, replacing the thermostat or water pump, and refilling the coolant.

Agent Prediction: The main causes of engine overheating are low coolant, coolant leaks, and old cooling parts. Typical fixes include repairing any leaks, replacing the thermostat or water pump, and refilling the coolant.

Judge Verdict: Well answered. The agent mapped 100% of the mechanical components and fluid issues required for the fix.

Question 6 : How can you recognize a bad clutch while driving?

Expected: You can recognize a bad clutch by hard shifting and noticing that the RPM rises without acceleration.

Agent Prediction: To recognize a bad clutch while driving, you may notice hard shifting and an RPM increase without any acceleration.

Judge Verdict: Well answered. Extracted the exact behavioral symptoms of a slipping clutch accurately.